CS-222-VSC: Python Programming
Lectures: 15 + Practical (Credits-2)

Prerequisites: None

Learning Objectives:

1. To introduce programming concepts using python.
2. To use various data structures and control structures python.
3. To learn use of built-in functions, modules and develop user defined functions.
4. To write and execute python programs using OOP.

Course Outcomes: On completion of this course, students will be able to:

CO1: Understand basic constructs of programming.

CO2: Determine the methods to create and develop Python programs by utilizing the data structures like lists, dictionaries, tuples and sets and control structures.

CO3: Use built-in functions, modules and develop user defined functions for problem solving.

CO4: To write python programs using OOP for problem solving.

Unit 1: Introduction to Python **02 (CO1)**

1.1 Need, Applications, program structure, interactive mode and script mode

1.2 Python Fundamentals – Keywords, Operators, Variables, Indentation, Comments, Input and Output in Python

Unit 2: Data Handling and Control statements **09 (CO2)**

2.1 Data Types – int, float, complex, boolean, none, string

2.2 String - declaration, manipulation, special operations, escape character, string formatting operator, Raw String, Unicode Strings, built-in String methods.

2.3 List - Concept, creating and accessing elements, updating & deleting elements of list, traversing a list, reverse, built-in List Operators - Concatenation, Repetition, In, Built-in List functions and methods.

2.4 Tuple - Accessing values in Tuples, Tuple Assignment, Tuples as return values, Variable- length argument tuples, Basic tuple operations - Concatenation, Repetition, in, Iteration, Built-in tuple functions, indexing, slicing and matrices.

2.5 Set - Concept, creating and accessing elements, Operations on Sets

2.6 Dictionary - Creating a Dictionary, Accessing Values in a dictionary, Updating Dictionary, Deleting Elements from Dictionary, Properties of Dictionary keys, Operations in Dictionary, Built-In Dictionary Functions, Built-in Dictionary Methods.

2.7 Conditional Statements - if, if-else, nested if-else

2.8 Loops - for, while, nested loops, loop control statements (break, continue, pass)

Unit 3: Functions and Modules **04 (CO3)**

3.1 Definitions and Uses, User defined functions, Function Calls

3.2 Types of functions, Parameters, Return values

3.3 Lambada functions

3.4 Recursion

3.5 Modules - Importing module, Creating & exploring modules, Math module, Pandas, Numpy

Reference Books:

1. An Introduction to Computer Science using Python 3 by Jason Montojo, Jennifer Campbell, Paul Gries, The pragmatic bookshelf-2013
2. James Payne, "Beginning Python": Using Python and Python 3.1, Wrox Publication
3. Introduction to Computer Science Using Python- Charles Dierbach, Wiley Publication Learning with Python ", Green Tea Press, 2002
4. Introduction to Problem Solving with Python by E balguruswamy, TMH publication-2016
5. Beginning Programming with Python for Dummies Paperback – 2015 by John PaulMueller

Practical Assignments of Python programming

1. Assignment on data types, operators
2. Assignment on control statements
3. Assignment on string, list, set, tuple, dictionary
4. Assignment on function
5. Assignment on Modules